README — ULCM_M1_Extraction_Prompt.md

Purpose: The runnable LLM prompt that operationalises the ULCM M1 extraction spec. One record in, one JSON object out, with verbatim spans backing every non-trivial field.

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1. What this document is

The **actual text** sent to the LLM for every included paper. It is a self-contained SYSTEM + USER prompt with:

- The extraction rules the LLM must obey (verbatim-span discipline, non-inference rule, cost-in-original-currency, primary-study-tier logic, FF-ROB and AMSTAR-2 propagation).
- The full JSON output schema — every field with its { "value", "span", "section" } shape and its response enum.
- The controlled vocabularies (from Appendix D.8 of the protocol) inlined so the LLM never needs a separate reference.
- The RQ → Decision Brief section mapping table.
- Three paraphrase variants (EX-1 / EX-2 / EX-3) for the k = 3 majority-vote workflow.
- The post-extraction span-validator prompt (Appendix F.5 of the protocol).

It is what an engineer wires into their pipeline. It is not a design doc — that is the extraction-fields spec (ULCM_M1_Extraction_Fields_by_RQ.docx).

2. How to run it — end-to-end

1. **Segment the paper.** Split the full text into labelled sections (Abstract, Methods, Participants, Intervention, Comparator, Outcomes, Results, Discussion, Cost if separate, and each Table N). This lets the LLM cite section IDs in spans.
2. **Substitute template variables:**
   - {{record_id}} — EPPI Study ID.
   - {{unit_of_extraction}} — "review" or "primary_study".
   - {{record_tier}} — "full" or "reduced" (primary studies only).
   - {{source_review_ids}} — sourcing-review sids (primary studies only).
   - {{rq_tags_hint}} — RQ(s) the record was screened into.
   - {{segmented_full_text}} — the segmented paper.
3. **Run k = 3 times** with paraphrase variants EX-1 (base), EX-2 (shuffled field order — anti-anchor), EX-3 (extract-or-justify-absence).
4. **Merge** per-field via majority vote. Any field where the three runs disagree is added to fields_flagged_for_human in the merged output.
5. **Validate** every non-null span through the span-validator prompt (§F.5). Any fail → human re-extraction.

6. **Human verify** all quantitative fields used in synthesis (100%) against the source PDF before any use downstream.
7. **Store** every raw response, prompt version, and model API string as JSONL under /llm_logs/ for audit reproducibility.

3. Model configuration (from protocol Appendix F.1)

Role
Primary extractor
Span validator

Replication candidate for both roles: latest OpenAI flagship. Both candidates locked at protocol sign-off; the exact model string is recorded in /llm_prompts/_meta.yaml and cited in the final report's deviation register.

4. What the JSON schema covers (at a glance)

The output object contains roughly **220 fields** (including two repeating arrays — additional_outcomes and rq_contributions), grouped as follows:

Group	Fields	Applies to
Study identification	sid, cit, doi, yr, lang, doctype	All records
Design & context	design, country, stream, geo_focus, setting, pop_*	All records
Severity	sev_strat, sev_scale, sev_cutoff, sev_quote	All records
Intervention ingredients	int_name, int_techniques (D.8.3a), int_adaptations (D.8.3b), int_theory	All records
Sub-modules *(v1.5 REVISION)*	int_submodules (keyed by programme), submodule_session_share_pct, submodule_contribution, submodule_evidence_type, submodule_manual_reference	Records where sub-module content is reported (IPT-G, PM+, HAP, THP, Friendship Bench, CBT-brief, PST, BA, MI, SSI, peer-support)
Dose	dose_n_sessions, dose_session_min, dose_total_contact_min, dose_homework_min, dose_band, dose_freq, dose_duration_wk, dose_delivered_pct	All records
Facilitator & delivery	fac_type, fac_train_h, fac_super, del_format,	All records

	del_modality, del_lang, group_size_*	
Primary effect	eff_metric, eff_value, **eff_se**, eff_ci_lo, eff_ci_hi, **eff_n_int**, **eff_n_ctrl**, eff_direction, eff_quote	All records
Durability (post / 3m / 6m / 12m)	eff_*smd, eff*_se, eff_*ci_lo, eff*_ci_hi, n_at_*int, n_at*_ctrl, remission_pct_*, trajectory_shape, sleeper_effect_flag, mechanism_claim, mediation_analysis	RQ8, RQ14 records; opportunistically elsewhere
Additional DEPRESSION outcomes (repeating array) *(v1.5 REVISION, v1.6 scope-constrained)*	additional_outcomes[]: outcome_id, outcome_construct (depression-only enum), outcome_scale (depression-instrument-only enum), outcome_scale_direction, outcome_is_primary, outcome_timepoint_weeks, outcome_timepoint_label, outcome_timepoint_verbatim, outcome_eff_metric, outcome_eff_value, outcome_eff_se, outcome_eff_ci_*, outcome_eff_p, outcome_n_int/ctrl, outcome_arm_int/ctrl_mean/sd, outcome_eff_direction, outcome_adjusted_flag, outcome_adjustment_covariates, outcome_subgroup, outcome_source_location, outcome_quote	Every {depression construct × instrument × time-point × subgroup} the study reports. No cap. Non-depression outcomes stay in their existing dedicated blocks (cost §3.7, engagement/safety §3.9, psychometrics §3.10, attrition §3A.4.3).
Engagement	engage_attendance_pct, engage_completion_pct, engage_dropout_pct, engage_reasons	RQ15 records; opportunistically
Safety	ae_reported, ae_type, ae_rate, safety_pathway, safety_monitoring_practice,	RQ17 records; opportunistically

	referral_pathway_type, step_up_trigger_ae	
Cost	cost_reported, cost_per_pt_local, cost_currency, cost_year, cost_perspective, cost_components, cost_metric, cost_driver, cost_effectiveness_ratio, ce_threshold_used, sensitivity_analysis_flag	RQ16 records; opportunistically
Psychometrics	psychom_instrument, psychom_setting, psychom_language, psychom_sensitivity, psychom_specificity, psychom_cutoff, psychom_reliability, psychom_construct	RQ18 records
RQ-specific driver / component / dose / spillover / stepped-care / trajectory fields	~50 fields tied to RQ1, RQ2, RQ3, RQ4, RQ5, RQ6, RQ7, RQ9, RQ10, RQ11, RQ12, RQ13, RQ14, RQ15, RQ17	Per-RQ
AMSTAR-2 checklist *(v1.4 REVISION)*	amstar2_mode, amstar2_i1…i16 (verdict + span + notes each), amstar2_critical_weaknesses, amstar2_noncritical_weaknesses_count, amstar2_band_derived, amstar2_band, amstar2_inherited, amstar2_inherited_source, amstar2_override_reason	Review-level records
Primary-study block *(v1.2 REVISION)*	record_tier, source_review_ids, linked_publications, rct_* quality items, attrition_, *trial_registered, trial_registry_id, funding_source, coi_*, n_enrolled_, *n_randomised_*, arm_definitions	Primary studies only

Fatal-Flaw ROB *(v1.3 REVISION)*	ffrob_coding_pair, ffrob_minutes, ffrob_c1a…c5b (verdict + span each), ffrob_c1_notes…c5_notes, ffrob_overall_decision, ffrob_failed_criteria	Primary studies only
Provenance & bookkeeping	rob_fatal_flaw, extractor_confidence, fields_flagged_for_human	All records
RQ tagging & report mapping	rq_tags, workstream, template_section, rq11_only	All records
Study-to-RQ contribution matrix (repeating array) *(v1.5 REVISION)*	rq_contributions[]: rq_id, rq_contribution_type, rq_contribution_summary, rq_contribution_strength, rq_contribution_direction, rq_contribution_data_fields, rq_contribution_template_section, rq_contribution_quote	One entry per RQ the study contributes to. Materialised as long-format row-per-{sid × rq_id} in the exported sheet.

Every field uses the uniform provenance shape { "value": …, "span": "<verbatim ≤ 40 words>", "section": "<Methods 2.1 | Table 3 | Results, para 4>" }.

5. Rules the prompt enforces

Twenty-six numbered rules. The most consequential:

- **Rule 1** — verbatim spans mandatory; ≤ 40 words; character-for-character copy; section reference required.
- **Rule 2** — cost figures in original currency and year; no pre-conversion.
- **Rule 2a** — SE extracted as reported; never back-calculated from CI or p-value at extraction time.
- **Rule 2b** — arm-level Ns per time-point are the analytic Ns, not enrolled Ns.
- **Rule 3** — provider perspective is the headline cost figure; societal recorded but not headline.
- **Rule 5** — no primary-ingredient assignment at extraction (that happens at synthesis).
- **Rules 11–16** *(REVISION)* — primary-study workflow: capped RQ scope, tier definitions, sourcing-review link, deduplication, RCT quality coding.
- **Rules 17–19** *(REVISION)* — FF-ROB checklist coding and propagation to rob_fatal_flaw; asymmetric consequence (excluded from quant, retained in narrative).
- **Rules 20–26** *(REVISION)* — AMSTAR-2 rapid critical-domains mode; per-item response grammar; confidence-band derivation from the 7 critical items; inherit-from-

umbrella shortcut; direct routing to the eight review-only RQs (RQ1, RQ2, RQ3, RQ4, RQ11, RQ15, RQ17, RQ18) where AMSTAR-2 is the sole quality signal.

- **Rules 27–29** *(REVISION)* — sub-module decomposition per named programme (IPT-G, PM+, HAP, THP, Friendship Bench, CBT-brief, PST, BA, MI, SSI, peer-support), with sub-module vocabulary keyed by programme; repeating additional_outcomes[] array for every {construct × instrument × time-point × subgroup} the study reports; repeating rq_contributions[] array — one entry per {study × RQ} — materialised as long-format row-per-{sid × rq_id} in the exported extraction sheet.

6. Paraphrase variants (k = 3 majority vote)

Variant	What differs	What it tests
EX-1	Base prompt as written.	Baseline extraction.
EX-2	Schema fields presented in shuffled order (permutation fixed per run, logged).	Anti-anchoring — does the LLM extract the same values regardless of field order?
EX-3	Adds per-field absence_notes map: for every null field, LLM records a ≤ 15-word reason.	Prevents “silent nulls” — the LLM must actively justify absence rather than default to null.

Merge rule per field:

- All three agree → take value; span from EX-1.
- Two of three agree → take majority; span from a majority run; add field ID to fields_flagged_for_human.
- All three disagree → set value = null; store candidates in /llm_logs/; flag for human.
- extractor_confidence = mean of the three per-run confidences.

7. Calibration before live use

Before running on the full corpus:

1. **Pilot on 5 studies** per protocol §4.4 (extraction template pilot).
2. **Human gold standard**: the two core team members (LS, SP) double-extract each pilot paper.
3. **Iterate the prompt** until per-field agreement with the human gold standard reaches an acceptable level. Log every prompt version in /llm_prompts/_meta.yaml.
4. **Only then deploy** on the full corpus. Every prompt change post-deployment is a protocol deviation and goes into the deviation register.

8. What breaks the pipeline (fail-fast conditions)

- Response is not valid JSON → LLM returned prose or a code fence. Re-run.
- Any span fails the verbatim-validator on 100% of quantitative fields → human re-extraction.

- extractor_confidence < 0.5 across > 30% of fields → the paper is likely poorly segmented; re-segment and re-run.
- ffrob_overall_decision = "High ROB — fatal flaw" AND record_tier was set to "full" → downgrade to "reduced" at merge; exclude from effect-size synthesis.
- amstar2_band_derived ≠ amstar2_band AND amstar2_override_reason is empty → flag for human adjudication.

9. Change log

Version	Date	Change
1.0	2026-07	Initial prompt derived from Protocol Appendix F.4 + RQ-specific fields per extraction spec.
1.1	2026-07	Added eff_se, arm Ns, per-time-point SE / CI / N.
1.2	2026-07	**REVISION** — primary-study workflow (rules 11–16); tier flag; sourcing-review link; RCT quality items; attrition; trial provenance; arm Ns.
1.3	2026-07	**REVISION** — Fatal-Flaw ROB checklist (rules 17–19): 5 criteria × 9 sub-items; propagation to rob_fatal_flaw.
1.4	2026-07	**REVISION** — AMSTAR-2 itemised (rules 20–26): 16 items with critical / non-critical split; rapid critical-domains mode; confidence-band derivation; tie-in to review-only RQs.
1.5	2026-07	**REVISION** — sub-modules (rule 27) — vocabulary keyed to 11 programme families with session-share % and contribution; multi-outcome (rule 28) — repeating additional_outcomes[] array; study-to-RQ matrix (rule 29) — repeating rq_contributions[] array

		with contribution type / strength / direction, materialised as long-format row-per-{sid × rq_id} in the export.
1.6	2026-07	Scope-narrowed additional_outcomes[] to depression outcomes only per review-team decision. outcome_construct enum restricted to five depression constructs; outcome_scale enum restricted to 20 validated depression instruments. Non-depression constructs (anxiety, functioning, wellbeing, QoL) explicitly excluded from this block. Rule 28 rewritten with the scope constraint.

Files: ULCM_M1_Extraction_Prompt.md · this README (README_Extraction_Prompt.md and .pdf). **Companion:** ULCM_M1_Extraction_Fields_by_RQ.docx and its README (README_Extraction_Fields.md). **Anchors:** ULCM Rapid Review Protocol v9 July (Appendix F.4 = prompt template; F.5 = span validator; F.6 = calibration).